

Readying for production

USER-ORIENTED DESIGN IN POWER BI

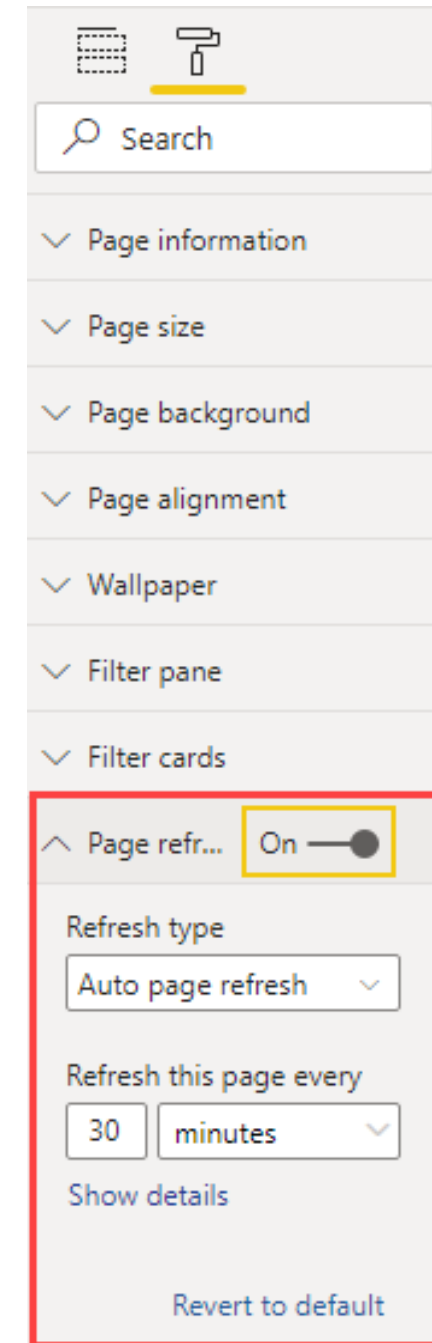


Maarten Van den Broeck

Content Developer at DataCamp

Automatic page refresh

- Controls how often a Power BI report will query for new data
- Only works with DirectQuery sources
- Fixed intervals allow for automatic refreshing at scheduled times



Change detection

- Works on tracking the aggregated value of a given field
- Can also use an existing calculated measure
- When there is a tracked change, Power BI automatically refreshes the data source and visuals using that source

Change detection

Improve performance by only refreshing your data if the result of this measure changes.

Measure type

☐ Existing

☒ New

Choose a calculation

Maximum

Choose a field to apply it to

Σ year

Check for changes every

5 Minutes

Seconds

Minutes

Hours

Fields

Search

> CPIPeriodicity

> CPISeasonality

▼ CPIData (2)

☐ area_code

☐ base_id

☐ begin_period

☐ Σ begin_year

☐ end_period

☐ Σ end_year

☐ item_code

☐ period

☐ periodicity_code

☐ seasonality_code

☐ series_id

☐ Σ value

☒ Σ year

Apply

Cancel

Performance considerations

- Refresh interval should be no less than the expected new data arrival interval
- Use Performance Analyzer to review the cost per query
- Each user with a report open runs independent refresh operations

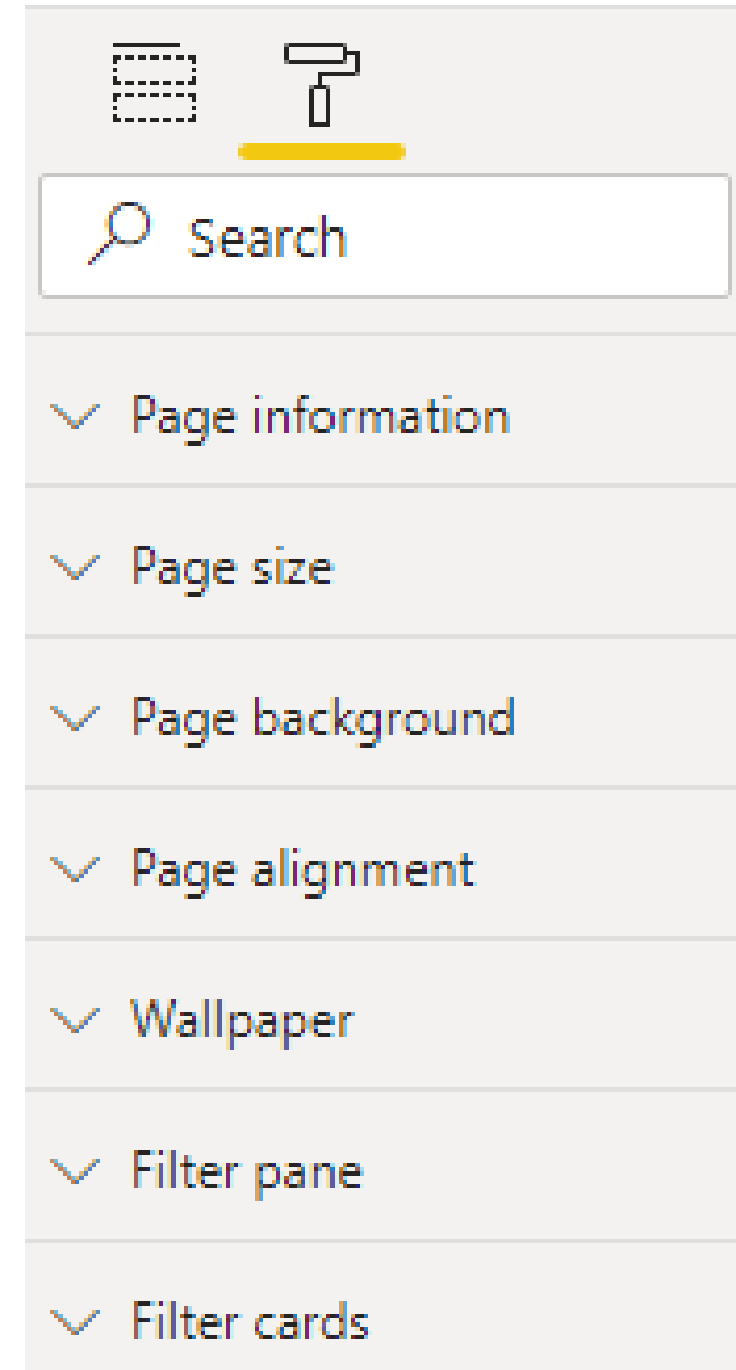
⊕ Change detection: Change detection Count (Disti...	86
⊖ Change detection: Change detection Count (Disti...	90
DAX query	47
Direct query	43
Other	43
📄 Copy query	
⊕ Total Quantity by Store	361
⊖ Products Sold	429
DAX query	137
Direct query	134
Visual display	17
Other	275
📄 Copy query	
⊕ Total # of Sales	282
⊕ Products Sold Over Time	487

Capacity requirements

- Change detection is only available in a Premium SKU
- Shared workspaces may perform scheduled refreshes at a minimum interval of 30 minutes
- Dedicated capacity allows you to update as frequently as once per second, depending on the storage mode
- A maximum of 10 models with a change detection measure are allowed in a Power BI tenant--in large enterprises with lots of report developers, think about scheduled intervals by default

Configuring report pages

- Specific to a page of a report--multiple pages may have different settings
- Control size, alignment, and layout of report pages
- Control backgrounds and wallpapers
- Control the way the Filter pane looks



Let's practice!

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Paginated reports

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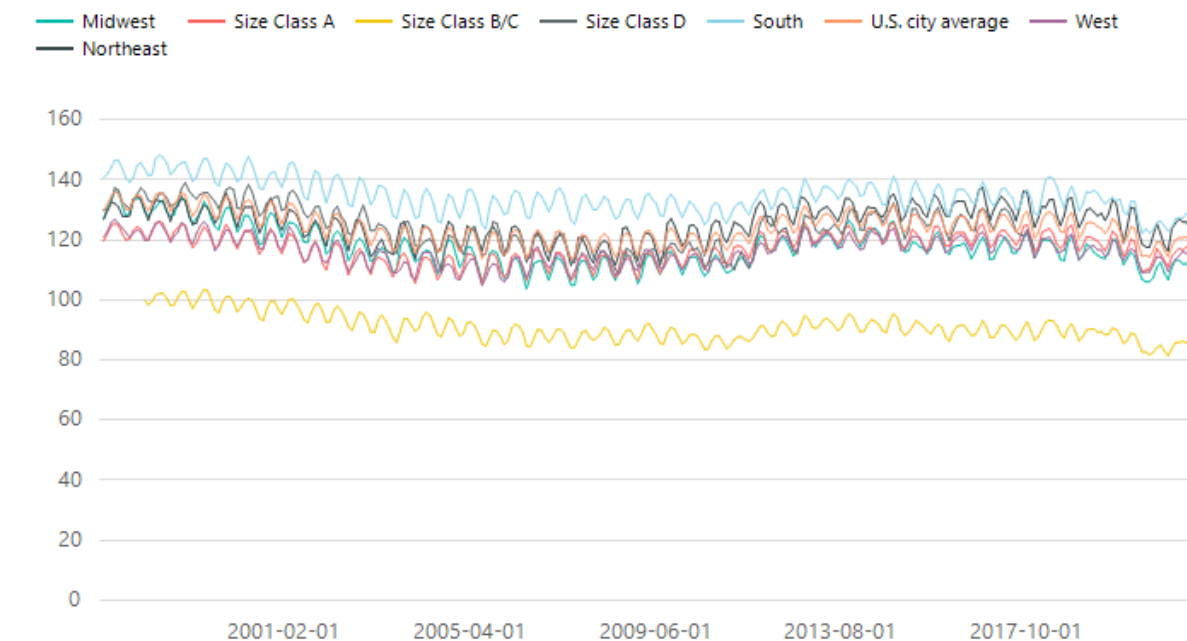
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What are paginated reports?

- "Pixel-perfect" report intended for the written page
- Created in the Power BI Report Builder

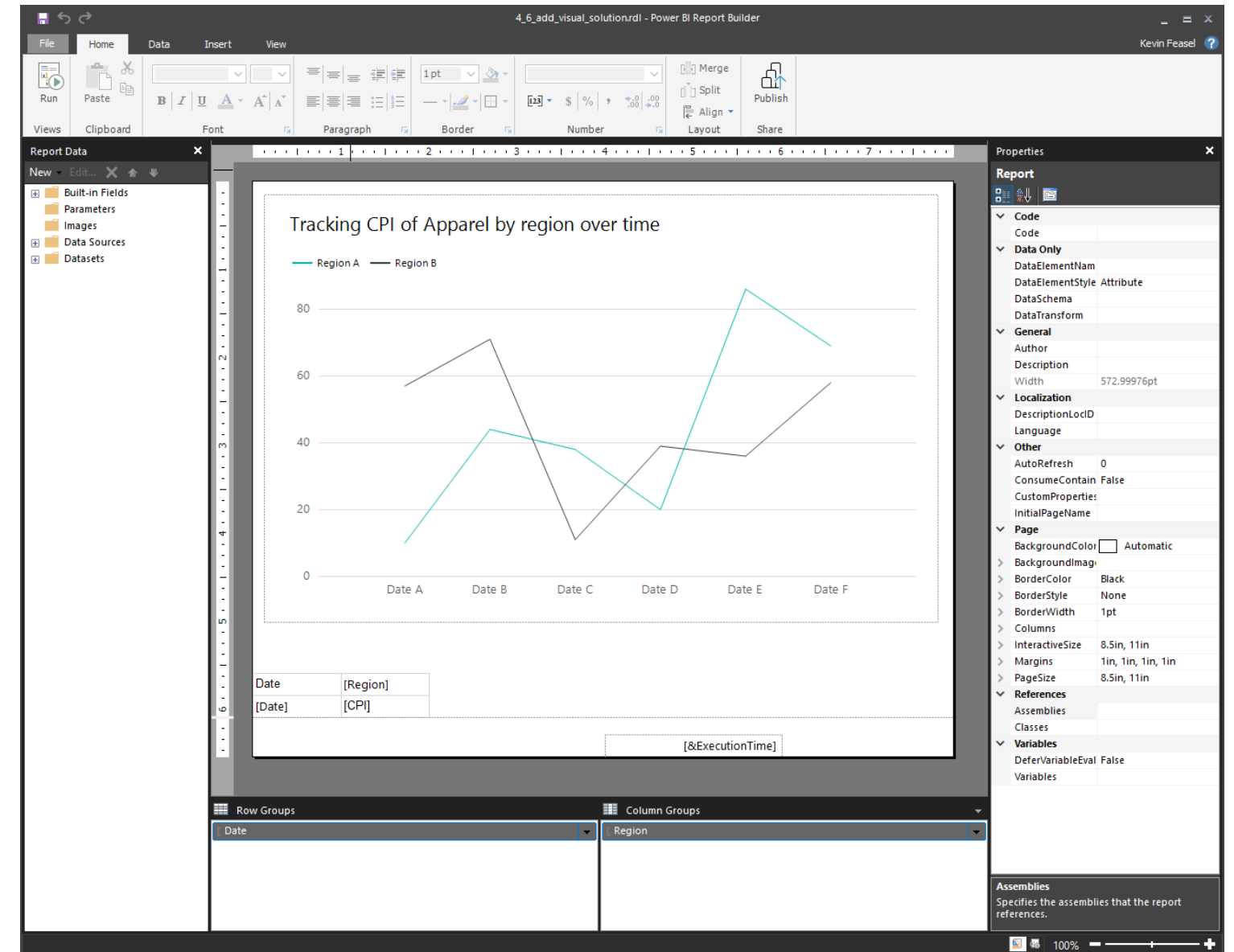
Tracking CPI of Apparel by region over time



Date	Midwest	Northeast	Size Class A	Size Class B/C	Size Class D	South	U.S. city average	West
1997-01-01	127.00	126.90	119.60		130.10	140.80	129.60	121.00
1997-02-01	130.70	129.80	122.20		129.80	141.90	131.90	122.80
1997-03-01	133.30	132.50	124.50		132.30	144.10	134.50	125.70
1997-04-01	136.40	132.10	125.50		137.30	146.60	136.10	126.80
1997-05-01	135.70	130.90	124.80		136.50	146.40	135.30	125.40
1997-06-01	131.40	128.00	122.00		132.60	143.70	132.40	123.90
1997-07-01	127.70	127.70	120.00		131.40	140.50	130.20	122.40
1997-08-01	129.20	128.10	120.30		130.40	139.10	130.00	120.80
1997-09-01	133.10	133.30	123.20		133.40	140.90	133.00	122.30
1997-10-01	133.80	134.60	124.40		135.20	144.80	134.90	123.30

The origin of paginated reports

- SQL Server Reporting Services (SSRS)
- SSRS Report Builder
- Power BI Report Builder



Comparing against dashboards

Power BI paginated reports

- Easily print out tables over multiple pages
- Dozens of visuals, including many classics (line chart, bar/column chart, scatter plot, box plot, radar chart, pie chart)
- No cross-filtering or slicing on reports--more for read-only

Power BI dashboard reports

- Limited print capabilities
- Dozens of visuals, but incomplete overlap!
- Cross-filtering and active slicing

Important limitations

- Viewing a report requires a Pro license, but saving a report requires Premium capacity
- No sharing data sources or datasets between paginated reports
- No drillthrough or click-through capabilities or linked reports
- Limited in-app interaction, but you can export it to Word, Excel, CSV, or PDF format.

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Creating paginated reports

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